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Child Gender and the Family

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Abstract and Keywords

Parents treat sons and daughters differently, but the causes and manifestations of the discrimination depend on context. This chapter reviews the literature on child gender in developing and developed countries and discusses methodological issues that arise when studying child gender effects. In many Asian countries, girls receive less adequate nutrition and health care, and sex-selective abortion is common. In the United States, parents of sons are more likely to be married than parents of daughters, and paternal labor supply and household expenditure patterns vary by child gender. In both cases, the patterns can be explained by differences in returns to children by gender or preferences for sons, and the responses to child gender are shaped by parents' relative bargaining power. The particular mechanisms depend on culture and constraints faced by parents. Methodological issues include endogenous fertility and child mortality in developing countries and endogenous family structure in the United States.

Keywords: gender, children, family, household, discrimination, bargaining, dowry, fertility, missing women

PARENTS treat sons and daughters differently, but the nature of the differences depends on context. In parts of Asia, for instance, sex-selective abortion is practiced, and the girls who are born receive less adequate nutrition and health care than their brothers. Differences are generally less stark in Western countries. They relate to the type and amount of interaction parents have with their children and the indirect effects of child gender on family structure. This chapter discusses the literature in economics that seeks to understand the causes and consequences of the differences in treatment.

How does economic theory explain why parents would treat sons and daughters differently? In a unitary model of the household, parents allocate goods and time to maximize their joint utility over the number or well-being of sons and daughters, as well as their own consumption (Rosenzweig and Schultz 1982; Behrman, Pollak, and Taubman 1982). Parental preferences exhibit *unequal concern* when the utility function is asymmetric with respect to child gender (Behrman et al. 1982). Constraints depend on child endowments, as well as prices and income. Some models introduce production functions for child well-being or human capital that differ by child gender. Nonunitary models allow mothers and fathers to have different preferences or constraints. Most notably, divorce threat models such as Manser and Brown (1980) and McElroy and Horney (1981) predict that increasing a parent's relative bargaining power will result in an allocation more favorable to him or her.

The specific model depends on context as preferences and constraints are shaped by culture, policy, and geography. For instance, gender specific agricultural production functions may be needed to explain discrimination against girls (p. 260) in rural areas of developing countries. In developed countries, where family structure depends on child gender, divorce and marriage should be treated as endogenous.[1] A common theme in the economics literature on child gender and the family is identifying the model—pure preferences, returns, or bargaining—that underlies the discrimination.

This chapter reviews several issues in the literature on child gender and the family. The second section pertains to developing countries with a focus on South Asia and China, and the third section pertains to the United States. The fourth section concludes.

Gender Bias in Developing Countries

One of the most striking manifestations of gender bias in the developing world is the phenomenon of missing girls in Asia (Sen 1990). Historically, these girls were likely to have been victims of infanticide or neglect (Rose 1999). In recent years girls are less likely to be born in the first place as access to ultrasound has enabled parents to engage in sex-selective abortion (Jayachandran 2016).

One way to detect the problem of missing girls is to compare reported sex ratios at birth —the number of boys born per hundred girls—to the biological norm of 105.5 (Johansson and Nygren 1991). In 1980, the ratios were 107 in China and 106.5 in India (Jayachandran 2016). The ratios skyrocketed in the 1980s as the advent of ultrasound technology allowed parents to engage in sex-selective abortion. In 2007 they were 117 in China and 110.8 in India (Jayachandran 2016). Bhalotra and Cochrane (2010) estimate that about 480,000 girls were selectively aborted annually in India between 1995 and 2005 alone, and Chen, Li, and Meng (2013) show that 40 to 50 percent of the increase in the sex ratio in China in the 1980s was attributable to the introduction of ultrasound.

Girls who are born and survive are treated more poorly than boys in many respects. They are less likely to receive medical care (Chen, Huq, and D'Souza 1981; Das Gupta 1987; Pande 2003; Boorah 2004; Oster 2009; Hazarika 2000; Kubo and Chaudhuri 2017). They are more poorly nourished (Chen et al. 1981; Bose 2011; Gittelsohn 1991). They are breastfed for shorter durations (Barcellos, Carvalho, and Lleras-Muney 2014; Jayachandran and Kuziemko 2011). And they receive less adult care time (Barcellos et al.

2014). As a result, girls' health outcomes are less favorable than boys' (Pande 2003; Barcellos et al. 2014).

This section will begin with a discussion of two theories of discrimination against girls—differences in returns, and parental bargaining power—and the evidence in support of them. It will proceed to discuss four methodological points that affect interpretation of empirical results.

(p. 261) Differences in the Net Returns to Sons and Daughters in India

Economists first addressed the issue of child gender in the context of rural areas of developing countries, where income derives mainly from agriculture. Boserup (1970) and Bardhan (1974) emphasize the role of climate and soil conditions in shaping differences in the treatment of boys and girls. The idea is that gender differences in survival depend on differences in productivity, which in turn depend on climate and soil conditions. Bardhan (1974) hypothesizes that excess female mortality rates will be lower in rainier rice-growing regions where women are relatively more prouctive. Rosenzweig and Schultz (1982) estimate a model in which gender differences in child survival depend upon rainfall-induced variation in productivity in India. Their findings are consistent with Bardhan's hypothesis.[2]

In a more recent paper, Carranza (2014) uses data on soil type in India to test whether differences in productivity affect child sex ratios. She notes that women are more productive in areas where soil is more loamy than clayey and uses soil type as an instrument for women's agricultural labor force participation in an equation explaining child sex ratios. She finds that child sex ratios are higher in areas of India where soil is more loamy and where women are more engaged in the labor market, and she concludes that increasing women's participation in the labor market could help normalize child sex ratios.

The dowry system provides an additional source of differences in (net) returns to sons and daughters. In some parts of India parents pay up to eight years of household income to marry off a girl, and dowry payments have been increasing over time (Rao 1993; Deolalikar and Rao 1998; Anderson 2007). As a result, the birth of a girl relative to a boy generates a negative income effect. If one margin on which parents respond to the negative income shock is to eliminate the costly daughter by committing infanticide or aborting her, we would expect to see a positive effect of dowry obligation on sex ratios. At first glance, the hypothesis would seem to be testable using a regression model of sex ratios on dowry payments. However, dowry is correlated with parents' unobservable preferences and constraints. To establish causality, it is necessary to identify a source of exogenous variation in dowry obligation.

Bhalotra, Chakravarty, and Gulesci (2016) note that an increase in the price of gold increases dowry obligation because gold is a large component of dowry. They exploit exogenous variation in the world price of gold to test whether dowry affects the survival of girls relative to boys. Using monthly time series data and an event study around the increase in the price of gold in international markets 1980,[3] they show that an increase in the price of gold increases excess female mortality. The effects are greatest in areas of India where son preference (proxied by population sex ratios) is stronger and greater for Hindus, who are more likely to practice dowry, than for Muslims. These results are consistent with a model in which the price of gold affects the net returns to having a daughter relative to a son, which in terms affects gender differences in survival.[4]

(p. 262) Deolalikar and Rose (1998) and Rose (2000) employ an indirect approach for testing whether parents respond to gender differences in returns. When parents are unconstrained in credit markets, they will respond immediately to the substantial increase in expected lifetime wealth resulting from the birth of a boy relative to a girl. There will be a positive income effect on consumption of normal goods and leisure and a negative effect on savings in response to the birth of a son rather than a daughter.

Deolalikar and Rose (1998) estimate the effect of child gender on household savings and expenditures in the child's first five years of life using the following equation:

$$OUTCOME_{it}=\alpha_i+\upsilon_{vt}+\sum_{k=0}^{4}\beta_k BOY_{i,t-k}+\sum_{k=0}^{4}\gamma_k GIRL_{i,t-k}+\epsilon_{it},$$

(1)

where $BOY_{i,t-k}$ and $GIRL_{i,t-k}$ refer to the birth of a boy or a girl to household *i*, *k* years prior to year *t* in which an outcome is realized. The gender shock for each year *k* subsequent to the birth is the difference $BOY_{i,t-k} - GIRL_{i,t-k}$, and the corresponding effect of that gender shock is $\delta_k=\beta_k-\gamma_k$. α_i and υ_{vt} are household and village × time fixed effects, respectively, and ε_{it} is an error term that is initially assumed to be independent and identically distributed. While the βs and the γs may be biased individually due to endogenous fertility, the bias differences out from the δs.[5]

Deolalikar and Rose (1998) find little evidence that the gender shock affects outcomes for landless and small farm households. However, there are notable findings for medium and large farm households, which are less likely to be constrained. Consistent with an income effect, overall consumption and expenditures on clothing, travel, electricity, water, and domestic labor increase more after the birth of a boy rather than a girl. Households spend more on medical care in the year of the birth and on fats and oils in the year of birth and in each of subsequent four years. These results suggest that boys receive more nutritious food and receive better medical care than girls, consistent with either preferences for or an income effect.[6]

Rose (2000), using the same empirical model, finds that in poorer households, women work less and men work more after the birth of a boy. In less poor households, both women and men work less after the birth of a boy relative to a girl. This implies that responses to child gender depend on households' access to credit markets.

Household Bargaining

Bargaining models provide an alternative to returns-based models in explaining the observed relationship between women's value in the labor market and girls' well-being. Bargaining models predict that, if mothers are less apt to discriminate against daughters, girls will do better in households in which mothers have greater control over resources. Two early pieces of evidence are Thomas (1994), who shows that in Ghana, Brazil, and the United States, boys and girls tend to be treated more equally when mothers are better (p. 263) educated or have more unearned income relative to their fathers, and Rose (1999), who shows that, in India, girls' mortality rates relative to boys' are greater in households where mothers' education is lower and fathers' education is higher.[7]

Regression-based estimates are biased when omitted variables—such as gender differences in returns to schooling or labor market opportunities—are correlated with parents' education or nonlabor income. Qian (2008) exploits a natural experiment from China to avert this bias. The post-Mao reforms of the 1980s led to an increase in the price of tea, which tends to be produced by women, and orchard crops, which tend to be produced by men. Women's relative income, and therefore bargaining power, increased in tea areas and decreased in orchard areas in response to the reforms. Qian finds that—holding spouses' income constant—girls' survival rates are higher when women's income is higher and men's income is lower. She also finds that increasing women's income increases schooling of both sons and daughters, while increasing men's income increases schooling of sons only. The asymmetry in the effects of the relative income shifts is consistent with a bargaining model.

Another set of bargaining models considers the effect of child gender on maternal well-being. Li and Wu (2011) predict that because mothers of boys are more valued than mothers of girls, they will have greater bargaining power. They find that Chinese mothers of sons have a greater role in household decision making, are better nourished, and have more favorable anthropometric outcomes than mothers of daughters. Using cross-country data for developing countries, Bhalotra et al. (2016) find that the presence of a son moderates impacts of unemployment on physical domestic violence. Milazzo (2014) finds that in India, mothers of sons have lower mortality rates.[8] Using data from Bangladesh, Heath and Tan (2016) find that mothers of daughters have greater autonomy and interpret autonomy as reflecting bargaining power. This suggests that mothers have relatively stronger preferences for spending on daughters and seek autonomy to obtain the resources for them. Baranov, Bhalotra, and Maselko's (2016) randomized controlled study from Pakistan shows that provision of mental health services to women suffering from maternal depression disproportionally benefits girls. This suggests that the negative effect of having a girl rather than a boy on women's status in the household can be mitigated by policy and that there are feedback effects on the well-being of their daughters.

Missing Girls: Empirical Issues

Several empirical issues arise when studying the phenomenon of missing girls.[9] First, parity matters. Later-born girls are far more likely to be missing than earlier-born girls. For instance, Das Gupta (1987) finds staggering gender differences in mortality rates for higher-parity children of educated mothers in rural Punjab, India—a state noted to have particularly strong child gender discrimination. While the under-5-years-of-age mortality rate of firstborn girls is 17 percent lower relative to that of firstborn boys, the mortality rate of girls of second parity or above is nearly 2.5 times as high as similar boys. (p. 264) Muhuri and Preston (1991) report that girls with older sisters are at a 5.8 times risk of dying than girls without older sisters in Bangladesh. As with mortality, the propensity to abort female fetuses increases dramatically with parity (Chen et al. 2013 for China; Bhalotra and Cochrane 2010 and Portner 2016 for India).

Second, the ability of parents to learn the sex of a child prenatally affects the interpretation of data on the well-being of girls born. For instance, sex-selective abortion allows parents to shift mortality from the post- to the prenatal period. The results on this point are mixed. Ankukriti, Bhalotra, and Tam (2016) find that introduction of ultrasound reduced neonatal mortality and postneonatal under-5 mortality in India. They estimate that for every five girls aborted, about one additional girl survives to age 5, and that access to ultrasound eliminated the postneonatal mortality differential for third-born children. However, Almond, Li, and Meng (2014) find that access to ultrasound was associated with *higher* neonatal female mortality in China between 1975 and 1992, and Hu and Schlosser (2015) find no effect from 1992 forward in India.

Another channel through which ultrasound affects the well-being of girls is prenatal investments. For instance, Indian mothers who learn that they are carrying girls are less likely to be vaccinated for tetanus than those who are carrying boys. As a result, newborn girls are far more likely to die from tetanus. Bharadwaj and Lakdawala (2013) calculate that between 2.6 and 7.2 percent of excess female neonatal mortality in India is attributable to women pregnant with daughters failing to receive a maternal tetanus vaccination.

Third, while the extent of a missing girls problem can be detected from aggregate data it is impossible to identify at an individual level. It would be necessary to have data on pregnancies prior to potential sex-selection abortions. This is why researchers typically define their dependent variables as sex ratios when estimating the determinants of sex-selection abortion and infanticide. For instance, Qian’s (2008) dependent variable is sex ratio by county and cohort.

Rose (1999) develops an alternative approach that allows her to avert the problem of underreporting of girls’ births using individual-level data. She estimates the following equation:

$$\Pr(Girl_{it} | Child_{it}) = \alpha + \theta' X_{it} + \epsilon_{it},$$

(2)

where Pr(Girlit|Childit) is the probability that a child born to mother *i* in year *t* is a girl, and Xit is a vector of regressors that includes rainfall shocks experienced by the household of mother *i* in year *t*. She finds that the estimated coefficient on rainfall shock in the year of birth is positive, and shows, using Bayes' rule, that this implies that the survival of girls born in low-rainfall years is lower than the survival of boys born in low-rainfall years. The effects are most pronounced for landless households that are more likely to be credit constrained. Because the study period is 1969–1971—well before the introduction of sex detection technology—the girls must have been lost to direct or indirect infanticide. The implication is that parents who are unable to smooth consumption cope with adverse shocks by sacrificing their daughters' survival.[10]

(p. 265) A caveat in both aggregate and individual studies aimed at estimating the determinants of missing girls is that the normal sex ratio at birth of 105.5 applies to women who are well nourished and healthy during pregnancy. However, because male fetuses are frailer than female fetuses, nutritionally deprived women are less likely to have sons. Johansson and Nygren (1991) report sex ratios of about 100—more than five points below the biological norm—when pregnant women are poorly nourished. Even when it is not possible to know whether the missingness arises from biology or behavior, it is possible to speculate. In the case of Rose (1999), we would expect girls born in adverse years to be more, rather than less, likely to survive. This further supports a behavioral rather than biological interpretation of the findings.[11]

Fourth, when there are missing girls, researchers need to be cautious in interpreting gender effects for children who are born and survive. Gender bias can be expressed along two margins, which are described in terms of the following equations:

$$\text{OUTCOMEi} = \alpha + \beta\text{GIRLi} + \gamma\text{Xi} + \varepsilon\text{i}$$

(3)

$$\text{Pr(Child i Survives)} = \text{a} + \text{bGIRLi} + \text{cZi} + \text{ui}$$

(4)

*OUTCOME*i is some measure of a child's well-being, *GIRL*i is a dummy variable indicating that child *i* is a girl, *Xi* and *Zi* are sets of observable child and/or parent characteristics, and *εi* and *ui* are error terms. Equation (3) describes the effect of child gender on the well-being of a surviving child, that is, along the intensive margin. Equation (4) describes the effect at the extensive margin of whether a child survives to the point at which the outcome is measured. When there is discrimination along the intensive margin, $\beta<0$. When there is discrimination along the extensive margin, $b<0$.

Issues arise when we consider Equations (3) and (4) together. The first issue is that there may be discrimination along one margin but not the other (Jayachandran 2016). For instance, some parents may have a greater propensity to abort a female fetus to avoid paying a dowry, suggesting that $b < 0$. However, given that the parents have a daughter, they may not discriminate along the intensive margin. In that case, β=0. Similarly, β=0 and b<0 when parents' utility functions exhibit unequal concern with respect to the quantity but equal concern with respect to the quality of their sons and daughters.

The second issue is selection bias. Equations (3) and (4) can be interpreted as a selection model in which Equation (3) is the main equation and Equation (4) is the selection equation. If corr (ε,u)≠0, estimates of β and γ are subject to selection bias. For instance, if β and **b** are negative, meaning there is discrimination along both margins, and the error terms are positively correlated, then an OLS estimate of β would be biased upward and the true extent of discrimination along the intensive margin would be understated. This would occur, for instance, if the error terms included unobservable health endowments that positively affect both survival and well-being. Intuitively, because girls (p. 266) are more positively selected, the population of surviving girls is hardier relative to the population of surviving boys.

At first glance it might seem that the matter of selection bias could be resolved by estimating Equations (3) and (4) as a selection model, as in Heckman (1979). But there are two challenges. First, Equation (4) requires data on the population of children potentially born. Second, to identify the model, there must be at least one variable in Z that is not included in X and not correlated with ε.

One way to investigate the link between sex-selective abortion and surviving girls' well-being is to estimate reduced-form models of the effect of ultrasound availability—which lowers the cost of eliminating an unwanted girl—on outcomes. Hu and Schlosser (2015) proxy availability of ultrasound with the state/year-level sex ratio at birth in India. They find that the ability to engage in sex-selective abortion leads to a reduction in girls' malnutrition but has no effect on boys' malnutrition.

Ankuriti et al. (2016) consider a natural experiment based on the timing of supply shifts due to the introduction of ultrasound in India and the relaxing trade laws that allowed for an increase in imports of ultrasound machines.[12] They exploit the empirical regularity that the sex ratio of first births is not abnormally high. Using a difference-in-difference-in-difference approach, they find that gender gaps in malnutrition, breastfeeding duration, and vaccination rates of later-born children narrowed in response to the supply shocks; differences in neonatal mortality rates narrowed; and differences in postneonatal mortality were completely eliminated.

Differential Stopping Behavior

Parents who seek to have a minimum or target number of sons are more likely to have a subsequent child after having a daughter. This son-preferring differential stopping behavior (SP-DSB) is a widespread phenomenon. It has been detected in Egypt (Yount, Langsten, and Hill 2000; Chakravarty 2015), Morocco (Ben-Porath and Welch 1976), Vietnam (Haughton and Haughton 1998), Korea (Park 1983; Arnold 1985; Larsen, Chung, and Das Gupta 1998), India (Jensen 2002; Arnold et al. 1998; Chaudhuri 2012; Barcellos et al. 2014; Anukriti et al. 2016; Portner 2016), and Bangladesh (Ben-Porath and Welch 1976; Chowdhury and Bairagi 1990).

One implication of SP-DSB is that girls will tend to be in larger families. Girls may fare worse than boys overall not because they are discriminated against relative to their brothers, but because they compete with more siblings for resources within the household. That is, there could be unequal outcomes across the population despite equal treatment within households.

Two studies use data from India to distinguish whether gender differences result from discrimination or family size. First, Jensen (2002), using fixed effects and instrumental variables techniques, finds that one-tenth to one-quarter of gender differences in educational attainment are attributable to SP-DSB. Second, Barcellos et al. (2014) note that (when gender at birth is random) differences in treatment of boys and girls (p. 267) before the parents have had time to have another child must be attributed to discrimination rather than family size. They compare health inputs and health outcomes of girls and boys within the first fifteen months of life. Relative to girls, boys get more child care time, are breastfed longer, get more vitamin supplementation, and realize more favorable anthropometric outcomes. The two studies together indicate that both discrimination and family size differentials contribute to the gender gap in child well-being.

Because breastfeeding inhibits fertility, mothers can reduce the likelihood of having a birth by extending breastfeeding duration. Using data from India, Jayachandran and Kuziemko (2011) find that the shortest durations are for girls and later-born children, with later-born girls weaned the soonest. This is a particular concern for households without piped water where alternatives to breastfeeding are unsanitary. The impact is substantial. Jayachandran and Kuziemko (2011) calculate that about 9 percent of excess deaths of Indian girls are attributable to differences in SP-DSP-induced breastfeeding duration differentials.

Considering Fertility and Mortality Jointly

There are two reasons that fertility decline would affect gender differences in mortality when parents have a preference for sons. First, there is a *parity effect*. Excess female mortality is concentrated at high parities, and fewer children means fewer children at high parities, where girls are most vulnerable. Second, there is an *intensification effect,* where lower fertility combined with son preference creates greater pressure on parents to have a boy at a lower parity. That is, they have less scope to “try again” for the boy. As a result, parents will be more apt to engage in infanticide or sex-selective abortion for earlier births. Some studies suggest that the intensification effect outweighs the parity effect. Das Gupta and Bhat (1997) find a negative correlation between sex ratios and fertility in Korea, China, and India, and Basu (1999) finds a similar correlation using cross-country data. Bhalotra and Cochrane (2010) find a negative correlation between stated preferences for desired number of children and the extent of sex-selective abortion in India.

The aforementioned papers are descriptive. Ebenstein (2010) exploits the One Child Policy implemented in China in 1979 as a natural experiment for estimating causal effects of fertility limits on sex ratios. In the early years of the policy, families who had more than one child were fined and denied services. There were reports of forced sterilization and abortions. The policy gradually relaxed so that families in rural areas were allowed a second child if their first was a girl. He uses variation in the strictness of the policy over time and across regions to show that stricter policy led to both lower fertility rates and higher sex ratios. That is, the combination of strong son preference and a severe limit on fertility motivated parents to engage in sex-selective abortion. Similarly, Anukriti et al. (2016) use exogenous variation in the availability of ultrasound technology to ask how the ability to limit fertility affects mortality. They show that improved (p. 268) access to ultrasound resulted in higher sex ratios along with lower fertility, with the largest increases stemming from families with first-born girls.

Jayachandran (2017) tests for an intensification effect using survey data from Haryana India. Her survey asks parents how many sons and daughters they would want their adolescent daughter to have, given she would have a specified total number of children. Parents overwhelmingly preferred for her to have a son if she could only have one child. The desired sex ratio falls to 120 if she could have two children and 112 if she could have three.

The results on the intensification effect have implications for our understanding of the relationship between economic development and female well-being. As development proceeds, women's education and labor market productivity increase. This increases the opportunity cost of childbearing and thus reduces the demand for children. This results in an intensification effect as parents feel a greater desire to have boys at lower parity, which in turn leads to an increase in the demand for sex-selective abortion. Paradoxically, economic development and expanded opportunities for women may exacerbate the phenomenon of missing girls (Bhalotra and Cochrane 2010).

Child Gender and the Family in the United States

Child gender matters in developed countries as well. However, the differentials are milder than in developing countries, and in some respects girls are advantaged. The most salient issues relate to family structure and the amount and type of interaction children have with their parents. This section will review findings in three branches of the empirical literature on child gender effects, and then describe how economists differentiate between preference and returns explanations for effects. It will emphasize economists' contributions, although results from the broader social science literature have bearing on interpretation of work in economics.[13]

Interaction with Parents

Parents respond to a child's gender from the day the child is born. Newborn boys tend to be described in masculine terms, such as "robust" and "strong," while newborn girls are more likely to be described as "delicate" or "soft" (Maccoby 1999). Sharper distinctions emerge within the first year. Parents tend to spend more time with daughters in teaching activities such as reading, telling stories, singing songs, drawing, and teaching new words and letters as early as 9 months of age (Baker and Milligan 2016) At 1 year, fathers spend more time singing and mothers spend more time reading to their daughters (Lundberg, McLanahan, and Rose 2005). Parents' greater propensity to engage in (p. 269) teaching activities with girls continues to kindergarten age, when parents spend more time reading to their daughters and taking them to concerts (Bertrand and Pan 2013).

Some differences may reflect biology rather than bias. For instance, girls may elicit more verbal interaction with parents than boys (Leaper, Anderson, and Sanders 1998; Leaper and Smith 2004). Girls exposed to more prenatal testosterone are more likely to develop typically male play styles (see, e.g., Auyeung et al. 2009). Within-sex correlation between hormone exposure and behavior suggests that differences in behavior between the sexes may also be linked to hormones (Bertrand 2010).

Fathers' interactions with their children are more sex specific, while mothers tend to treat boys and girls more equally. Fathers encourage their boys to engage in masculine behaviors and physical play, and they are more apt to discipline their sons than their daughters. Daughters are treated more gently (Maccoby 1999). Fathers spend more time with sons than with daughters (Yeung et al. 2001; Lundberg, Pabilonia, and Ward-Batts 2007; Mammen 2011) and more time overall with their children if they have a son (Baruch and Barnett 1986). The effects depends on the age of the child. Fathers spend 9 percent more time with sons between ages 3 and 5, but there is no difference at earlier ages (Baker and Milligan 2016). Fathers of older boys are more apt to participate in schoolwork and other activities than fathers of girls (Lamb et al. 1987; Morgan, Lye, and Condran 1988; Maccoby 1999).

Marriage, Divorce, and Fertility

Child sex affects family structure. Marriages with sons are less likely to end in divorce (Morgan, Lye and Condron 1988; Dahl and Moretti 2008; Bedard and Deschenes 2005), although the relationship has been attenuated in recent decades (Pollard and Morgan 2002).[14] Mothers whose first child is born out of wedlock are more likely to marry—to the father when the child is young and to someone other than the father when the child is aged 8 to 12 (Lundberg and Rose 2003)—if the child is a son. Shotgun marriage is more common for parents of boys (Dahl and Moretti 2008). Parents of boys report being happier in their marriages (Baruch and Barnett 1986; Katzev, Warner, and Acock 1994;

Cox et al. 1999; Mizell and Steelman 2000). As a result, girls are less likely to reap the benefits of living in a well-functioning two-parent family.

There is some evidence that fathers who are not married to a child's mother are more engaged with the child if it is a boy. For instance, boys born out of wedlock are more likely than girls to be given their father's surname (Lundberg, McLanahan, and Rose 2007). However, there is no evidence that child support payments depend on the child's gender (Lundberg et al. 2007; Mammen 2008).

The evidence on differential stopping behavior in developed countries is mixed. Dahl and Moretti (2008) and Ichino, Lidstrom, and Viviano (2014) find that married mothers are more likely to proceed to have a second child if the first is a girl. However, parents of sons are more likely to be married. This generates a positive marriage selection effect that outweighs the within-marriage effect, so, on net, women are more likely to have a second child when the first is a son (Ichino et al. 2014).[15]

(p. 270) Parents and the Labor Market

Sons and daughters spur parents into different roles in the labor market and the family. Lundberg and Rose (1999) interpret their finding that parents' earnings diverge more if they have a son relative to a daughter as evidence of greater specialization in parents' roles in families with sons. Lundberg and Rose (2002) focus on fathers' labor market outcomes. Their sample of white men born between 1947 and 1974 is disaggregated into men born before and after 1950. They consider four specifications of the gender of the man's offspring: the number of sons and daughters, whether there is at least one son or daughter, the gender of his first child, and the number of girls and boys by parity.

For the early cohort, each specification indicates that fathers of boys earn more per hour than fathers of girls. For instance, men in the early cohort whose first child is a son earn approximately 5.3 percent more per hour than those whose first child is a daughter. There are quantitatively and statistically significant effects of child gender on hours worked for the later cohort. For instance, men work sixty-nine hours more per year if their first child is a son. That men's labor supply and home production time increase in response to the birth of a son suggests that fathers of boys consume less leisure. Lundberg and Rose (2002) find no significant effect of child gender on mothers' labor supply.

Preferences, Returns, and the Value of Marriage

Why do families respond differently to sons and daughters? The two leading explanations are differences in preferences and differences in returns to investments in sons and daughters. The task of distinguishing between alternative models is more challenging for developed than developing countries. Researchers studying developed countries rarely have the luxury of natural experiments such as variation in weather or commodity prices that make it possible to test hypotheses about the effect of changing a parameter on children's outcomes.

One approach to distinguishing between alternative models uses survey data. A 2000 Gallup poll, for instance, asked men and women whether they would prefer to have exactly one son or exactly one daughter. Men were nearly 30 percentage points more likely to want a son and women were 5 percentage points more likely to want a daughter (Dahl and Moretti 2008). Baker and Milligan (2016) infer preferences from parents' ex post responses to questions about wantedness. Mothers were more likely to state, after a child was born, that the child was wanted if it was a girl rather than a boy. Otherwise, Baker and Milligan found no evidence of differences in preferences toward boys and girls. These survey results suggest that fathers prefer to have sons and perhaps that mothers prefer to have daughters. However, they do not necessarily imply a pure preference explanation as parents' responses to survey questions take into account differences in the returns to the children and unequal concern.

(p. 271) A second approach to distinguishing among explanations is to view alternative models in terms of the constellation of empirical results in the literature. Key findings discussed earlier are that fathers devote more time to families when they have sons, that fathers and mothers engage differently with their sons and daughters, and that parents' relationships are happier, more stable, and more traditional if they have sons rather than daughters.

The higher marriage rates and lower divorce rates of parents of sons indicate that there is greater marital surplus in marriages with sons relative to daughters (Lundberg and Rose 2003; Lundberg 2005). Parents with greater confidence in the success of their marriage will have greater incentive to engage in specialization within the household and to invest in household public goods. These predictions are consistent with Lundberg and Rose (1999), who find greater specialization between the home and labor market in marriages with sons, and Lundberg and Rose (2004), who find that households with sons spend more on housing.

But why would sons generate greater marital surplus than daughters? In a pure preference model in which fathers prefer sons to daughters and mothers have equal concern for sons and daughters, fathers have more to lose from divorce and therefore a lower threat point than fathers of daughters. As a result, the within-marriage allocation will tend to be more favorable to mothers of sons, with greater consumption of adult female goods and lower consumption of adult male goods (Lundberg 2005). Two pieces of evidence support this explanation. First, as discussed earlier, fathers of sons consume less of the private good of leisure. Second, consumption of personal care services, an adult female good, is greater in families with sons (Lundberg and Rose 2004).

In a returns model, fathers have a comparative advantage in parenting sons relative to daughters. They are role models and better positioned to teach gender-specific skills when they are married to the mother. In this case, mothers of sons will have lower threat points than mothers of daughters, and the outcome will be more favorable to fathers of sons relative to fathers of daughters (Lundberg 2005). The results on fathers' leisure and personal care—private adult male and female goods, respectively—are inconsistent with this prediction.

So far, this chapter has considered cases in which girls are the disadvantaged sex. In recent years, however, there is concern about boys and men falling behind. While men are more likely to reach the very top levels of education and occupations, the median man has been increasingly faring worse than the median woman in the labor market (Autor and Wasserman 2013). There is also a widening in the gap in college attendance favoring women. In 1960, 1.6 men graduated from college for every woman. By 2003, 1.35 *women* graduated college for every *man* (Goldin, Katz, and Kuzimenko 2006).

The deterioration in men's achievement over the last several decades may be explained by changes in their childhood experiences. Several studies show that boys are more adversely affected by fathers' absence.[16] Bertrand and Pan (2013) find that by kindergarten, boys are more likely than girls to misbehave. The differential increases with age and is greatest for children from disadvantaged and single-parent homes. Autor et al. (2016) also find that boys are less likely than their sisters to be ready for kindergarten. (p. 272) They have higher absence and suspension rates in grades three to eight, and the gaps are greatest for children from lower socioeconomic groups and nonintact households. These two studies suggest that paternal contact is more productive in the production of noncognitive skills of sons relative to their daughters. As a result, the increase in the prevalence of single-parent households has taken a greater toll on boys.

A concern with the productivity explanation is that the effect of father absence may be attributable to resources rather than contact. Autor and Wasserman (2013) cite results from the Moving to Opportunity (MTO) study—in particular Clampet-Lundquist et al. (2011)—as evidence that it is contact that matters. MTO randomly assigned some families from inner-city neighborhoods to more advantaged neighborhoods outside the city. We would expect the treatment to have had a favorable impact on both boys and girls. However, while girls who moved fared better, boys, surprisingly, fared worse.[17]

The greater sensitivity of boys relative to girls to father absence suggests a second explanation for the greater success of marriages of parents of boys relative to parents of girls. Because children are far more likely to live with single mothers than single fathers, marriage means children receive more paternal inputs into the production of child quality. The returns to these inputs are greater for boys than for girls. This results in greater marital surplus, higher marriage rates, and lower divorce rates for parents of boys. Both preferences and returns play a role in shaping parents' responses to child gender in the United States.

Conclusion

Parents respond to the gender of their children, but the reasons and manifestations vary by context. In China and South Asia, discrimination against girls in nutrition and health care, and the gender gap in the propensity to be born and survive, can be attributed to differences in the returns to sons and daughters as well as preferences. The extent of discrimination depends on parents' bargaining power. In the United States, fathers of sons are more engaged with their families than fathers of daughters. Boys live in more stable and traditional families with greater resources and greater marital surplus. There is evidence that fathers prefer sons over daughters and that the returns to paternal presence is greater for couples with sons.

Context matters for both methodology and theory. In Asia, where sex-selective abortion is common and mothers' fertility-stopping behavior depends on the sex of their children, endogenous fertility and gender are important considerations. In the United States, where divorce and nonmarriage are common, the endogeneity of marriage is key.

One set of findings at the intersection of the Asian and US contexts illustrates that it is culture rather than geography that matters. Several studies show that child sex ratios for Asian immigrant families are more similar to the ratios in their countries (p. 273) of origin rather than the US population as a whole. Using data from the 2000 census, Almond and Edlund (2008) find that sex ratios are abnormally high for higher-parity children of Indian, Chinese, and Korean parents. In particular, if the first two children were girls, there were 50 percent more sons than daughters at the third birth. Abrevaya (2009) reports similar findings using total US and California natality data and census data. He estimates that sex-selective abortion in the United States resulted in over two thousand missing Chinese and Indian girls between 1991 and 2004.

Some US findings carry over to other developed countries and others do not. On one hand, similar to the United States, Ichino et al. (2014) find daughter-preferring differential stopping behavior in Sweden and Italy, and Karbownik and Myck (2017) find, for Poland, that having a son increases family spending on the adult female good of women's clothing. On the other hand, Brown and Bhalotra (2016) find that fathers in the United Kingdom work significantly *less* following the birth of a son than they do after the birth of a daughter. Andersson et al. (2005) find no evidence of differential stopping behavior for first birth and daughter preference for higher-parity births among Danish, Norwegian, and Swedish parents. Differences across the developed world might be explained by differences in family law, leave policies, labor market structure, norms, or some other factor. Further research can help shed light on the reasons that child gender differences vary across contexts.

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Notes:

(1.) Anukriti and Dasgupta (this volume) have some discussion of the role of son preference in marriage in a developing country context.

(2.) See Giuliano (this volume) for a discussion of historical determinants of contemporary differences in gender roles.

(3.) The world price of gold increased in response to the Soviet invasion of Afghanistan and the rise of Khomeni to power in Iran.

(4.) This is also consistent with an income effect if an increase in the price of gold makes households poorer overall and girls are a normal good.

(5.) When parents engage in sex-selective abortion, child gender at birth is endogenous. This is not a concern in Deolalikar and Rose (1998) or Rose (2000) as they use data from the 1970s, before ultrasound was available.

(6.) The expenditure equations resemble Deaton (1989) and Subramanian and Deaton (1991), who test for gender bias by estimating effects of the age × sex composition of a household on expenditure patterns. These papers assume that total expenditures in each period are fixed.

(7.) See Heath and Jayachandran (this volume) for more discussion of the relationship between mothers' education and infant mortality.

(8.) It is possible that these findings are attributable to an overall household income effect from the birth of a son (Deolalikar and Rose 1998; Rose 2000). This is more likely to be the case in India, where boys generate a large income effect relative to girls, than in China.

(9.) Portner (this volume) discusses fertility in developing countries in light of son preference.

(10.) Many other papers show that discrimination against girls varies with transitory conditions. Behrman (1988) shows that Indian girls receive poorer nutrition than boys in the lean, but not the peak, agriculture season. Bengtsson (2010) finds that girls' body weight is more responsive to weather-induced income shocks in Tanzania. Björkman-Nyqvist (2013) finds that negative deviations in rainfall from the long-term mean have negative and highly significant effects on female enrollment in primary schools. The effect increases with the child's age. Bandara, Dehejia, and Lavie-Rouse (2015) find that girls' enrollment is more responsive to crop shocks in Tanzania. Singh and Shemyakina (2016) find that educational expenditures were more responsive to conflict for boys versus girls in Punjab insurgency. Maccini and Yang (2009) find that adverse early life rainfall shocks have long-term consequences for women.

(11.) A similar caveat that women who are better positioned overall applies to the broader literature on child gender is suggested by Trivers and Willard (1973). Empirically, younger and more educated women, women who are living with a spouse or partner, and women with greater cognitive ability are more likely to have sons (Norberg 2004; Almond and Edlund 2007; Dama 2013).

(12.) They implement the natural experiment by defining three periods reflecting the extent of ultrasound use and policy changes: a preultrasound period of 1973–1984, an early diffusion period of 1985–1994, and a late diffusion period of 1995–2005 The specific breakpoints were identified in Bhalotra and Cochrane (2010) through nonparametric analysis of aggregate data on ultrasound use and number of ultrasound scanners imported to India.

(13.) Raley and Bianchi (2006) provide a thorough review of the sociology literature.

(14.) See Lehrer and Son (this volume) for a discussion of economic models of divorce.

(15.) Adser à and Ferrer (this volume) discuss fertility in developed countries in more detail.

(16.) McLanahan and Sandefur (1994) document substantial gaps overall in the well-being of children from single-parent relative to intact families.

(17.) Some part of the finding that boys are more sensitive to paternal absence may be due to selection bias, as unmarried parents of sons are on average from lower-quality relationships than parents of daughters who divorce.

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